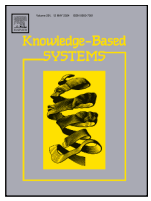
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LQMF-RD: A lightweight quantum-driven multi-modal fusion framework for rumor detection

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ABSTRACT

In recent years, automated rumor detection has garnered significant attention. Despite notable progress in multi-modal modeling for social media rumor detection, two major challenges remain: (1) the dynamic characteristics of social networks during the propagation process are often overlooked; (2) multi-modal features (such as text, images, and propagation graphs), are often poorly aligned and lead to redundant model parameters. To address these issues, we propose LQMF-RD, a lightweight quantum-driven multi-modal feature fusion framework for rumor detection. First, to capture the dynamic nature of rumor propagation, we design a Dynamic Graph Network (DGN) that leverages the spatiotemporal characteristics of propagation graph, effectively capturing both neighborhood dependencies and temporal evolution among nodes. Then, we employ amplitude encoding to project the extracted multi-modal features into a $[\log_2 N]$-dimensional quantum state space. Finally, we construct a Lightweight Quantum-driven Multi-modal Fusion Network (LQMFN), which enables deep interaction and fusion of multi-modal features through quantum convolution and pooling operations. LQMFN updates only 0.01M parameters, substantially reducing computational complexity and storage overhead. Experimental results show that LQMF-RD not only delivers superior performance on rumor detection tasks, but also achieves high computational efficiency and strong robustness to quantum noise.

1. Introduction

With the rapid evolution of online social media, posts have increasingly shifted toward multi-modal content, incorporating text, images, and more. Prior studies indicate that tweets with pictures are retweeted 11 times more than pure text ones [1]. However, this convenience and efficiency of propagation also enable unverified rumors to spread with comparable speed and reach. The spread of rumors is exceptionally rapid. If not identified and curtailed in time, such rumors can inflict immeasurable harm on public safety [2]. Therefore, developing effective and timely methods to detect and suppress rumors, thereby preserving a healthy online public discourse, has become an urgent challenge [3].

Traditional rumor detection methods primarily rely on unimodal information, focusing either on the content features of tweets [4,5] or on their propagation structures [6–9]. These methods typically employ deep learning models to analyze linguistic cues or to simulate user interactions within social networks (e.g., likes, comments, reposts), thereby capturing both content characteristics and propagation patterns. With the rise of multi-modal learning, researchers have sought to integrate diverse features, such as text, images, and propagation networks, through various fusion mechanisms to improve detection performance [10–13]. For instance, Khattar et al. [10] proposed a Multi-modal Variational Autoencoder (MVAE) to learn latent representations of multimedia posts for rumor classification, while Xiong et al. [12] combined a pre-trained BERT model for text feature extraction with a VGG-19 model pre-trained on ImageNet for image feature extraction, enabling joint multi-modal learning.

However, existing approaches still suffer from several critical limitations: (1) Most methods overlook the social context (e.g., reposts and replies), leading to incomplete modeling of posts. Although some studies attempt to enrich inputs by constructing heterogeneous graphs based on social context, such graphs are typically static and fail to capture the dynamic evolution of posts over time, despite temporal cues being essential for rumor detection; (2) In social graphs, repost and reply nodes are usually linked directly to the source post, disregarding the complex relationships and structural features embedded within the propagation process; (3) Current multi-modal fusion methods are still computationally intensive, hindering efficient deep feature fusion. To address these issues, we propose a Lightweight Quantum-driven Multi-modal Feature Fusion Framework (LQMF-RD) for rumor detection on

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0950-7051/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

social media. Specifically, the model first employs a Dynamic Graph Network (DGN) to effectively capture spatiotemporal evolution features from the propagation network graph. Then, a Lightweight Quantum-driven Multi-modal Fusion Network (LQMFN) is employed to achieve efficient interaction and deep fusion of multi-modal information. The main contributions of this paper are as follows:

- We propose LQMF-RD, which leverages quantum computing advantages to efficiently integrate multi-modal features such as text, images, and propagation network graphs.
- We design a Dynamic Graph Network (DGN) that combines structural and temporal attention mechanisms to effectively extract spatiotemporal features from the propagation network, capturing the evolution patterns of nodes across different time steps.
- We design a Lightweight Quantum-driven Multi-modal Fusion Network (LQMFN), which achieves deep feature interaction and fusion through quantum convolution and pooling operations. The network requires updating only 0.01M parameters, significantly reducing computational complexity and storage requirements.
- LQMF-RD achieves accuracy rates of 92.20% and 90.45% on the Weibo and PHEME datasets, respectively, outperforming existing baseline models. Further studies show that LQMF-RD not only provides superior computational efficiency but also exhibits strong robustness against quantum noise.

This paper is organized as follows: Section 2 reviews the related works. Section 3 proposes the LQMF-RD model. Section 4 sets up experiments and performs result comparison analysis. Section 5 gives the conclusion.

2. Related work

2.1. Rumor detection

In existing research, rumor detection is typically formulated as a binary classification problem. Early methods primarily relied on extensive hand-crafted features, aiming to model textual content, visual features, or propagation structures for training classification models [9,14–16]. For example, MA et al. [4] proposed a rumor detection model based on recurrent neural networks (RNN) to learn the sequential representation of events on Weibo. To capture more fine-grained sentiment features, FakeFlow [17] extracted sentiment words from the text and combined them with topic words to enrich feature representation. Regarding visual features, Jin et al. [18] found significant content differences between rumor-related and non-rumor-related images. Qi et al. [16] further fused frequency-domain and pixel-domain information to enhance the expressive power of image features. In terms of propagation structure, Liu et al. [9] employed a graph convolutional network (GCN) to extract the propagation structure of Weibo events, effectively modeling the rumor diffusion process and obtaining more discriminative structural representations. However, these methods mostly focus on single modalities (e.g., text, images, or propagation structure), whereas rumor information on social media is often inherently multimodal.

Single-modal methods struggle to fully capture the intrinsic relationships among heterogeneous information sources [19–23], which has prompted researchers to explore multimodal rumor detection approaches. Yang et al. [24] proposed a dual-stream fusion network based on deep learning, which integrates textual and visual features using a multi-head self-attention mechanism, mitigating issues such as cross-lingual differences, data imbalance, and multimodal fusion challenges in rumor detection. Zhuang et al. [25] proposed DyCR-Net, which achieves multi-granularity information fusion by dynamically allocating weights for the cross-modal routing network. Yu et al. [26] designed a BERT-based multi-domain multimodal fusion framework, which combines text embeddings obtained via BERT and image features extracted by VGG-19, and improves detection performance through a multi-domain classifier. To reduce reliance on labeled data, Zhang et al. [27] introduced a knowledge-enhanced multimodal pre-training framework, which jointly learns single-modal, cross-modal, and knowledge graph-based tasks in an unsupervised manner to obtain more robust multimodal feature representations. Huang et al. [28] initialized nodes with BERT embeddings and constructed two dynamic graphs representing information propagation and user interactions. After encoding each graph layer-by-layer with dual GCNs, a cross-attention temporal fusion unit integrated the spatiotemporal features.

Additionally, Jiang et al. [29] proposed a cross-modal enhancement method that transforms an n-shot classification problem into a more robust (n + z)-shot setting and employs a simplified linear probe to improve performance in few-shot multimodal rumor detection tasks. Yu et al. [30] proposed a multimodal consistency and inconsistency balancing network based on semantic relations, which performs deep interactions between text and image features under a contrastive learning framework, balancing cross-modal consistency and inconsistency to enhance detection accuracy. To address the challenge of detecting deeply mismatched rumors, Huang et al. [31] proposed a novel multimodal rumor detection framework, DEETSA, which integrates external dual-evidence enhancement with internal text-image similarity awareness. To further reduce noise interference, Jiang et al. [32] proposed the SAMPLE framework, which combines prompt learning with perceptual similarity mechanisms, effectively alleviating noise in multimodal fusion and achieving superior performance across various multimodal rumor detection tasks.

Despite the notable progress of multimodal fusion methods, most existing studies overlook the structural characteristics and dynamic evolution of rumor propagation in social networks, both of which are crucial for effective rumor detection [33]. Moreover, most multimodal fusion techniques commonly rely on cross-modal attention mechanisms, resulting in high computational and parameter overhead, thereby limiting their scalability in real-world applications. In contrast, our proposed method efficiently captures the deep correlations among text, images, and propagation graphs, while significantly reducing computational costs through a lightweight design, and more accurately models the temporal dynamics of rumor propagation.

2.2. Quantum neural networks

Quantum computing has achieved notable progress, especially in the application of quantum neural networks (QNNs). QNNs combine the principles of classical neural networks with the advantages of quantum computing, offering new possibilities for addressing complex problems. Narottama and Shin [34] introduced a reinforcement learning-inspired quantum neural network (RL-QNN), aimed at improving the efficiency of resource allocation in wireless communication. Li et al. [35] designed a QCNN that omitted pooling layers, which demonstrated its effectiveness in image recognition tasks. Lloyd and Weedbrook [36] explored the applications of quantum generative adversarial networks (QGANs) in various scenarios, highlighting their potential exponential performance advantage with high-dimensional training data. Huggins et al. [37] proposed quantum tensor neural networks (QTNNs), which consisted of parameterized unitary gates with a fixed tree-like network structure, suitable for handling classification and generative tasks, and exhibited great potential in dealing with random initial parameter issues and noise resistance. Yang et al. [38] used QCNNs for processing distributed feature extraction to explore confidentiality protection in voice recognition. Qu et al. [39] introduced a quantum neural network-based multi-modal fusion system (QNMF) for medical diagnosis. This system can process multi-modal medical data, enhancing the performance of intelligent diagnosis through data fusion from different modalities.

These studies demonstrate that QNNs can effectively handle high-dimensional data and achieve accelerated computation when processing large-scale datasets. Based on this, we propose a lightweight quantum-driven multi-modal fusion framework for rumor detection that leverages the advantages of QNNs.

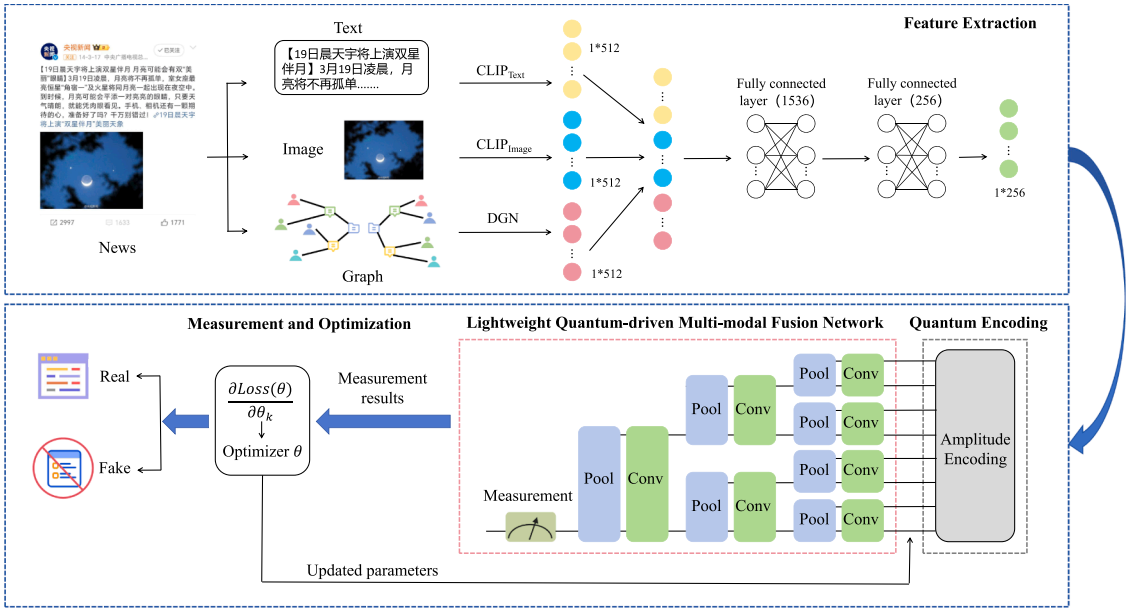


Fig. 1. The structure of the proposed Lightweight Quantum-driven Multi-modal Fusion Framework (LQMF-RD).

3. Lightweight quantum-driven multi-modal fusion framework

To effectively capture and integrate multi-modal information and leverage the advantages of quantum computing to enhance feature representation, we design a Lightweight Quantum-driven Multi-modal Feature Fusion Framework for Rumor Detection (LQMF-RD). As shown in Fig. 1, LQMF-RD consists of four modules: Feature Extraction, Quantum Encoding, Lightweight Quantum-driven Multi-modal Fusion Network(LQMFN), and Quantum Measurement and Optimization. Specifically, Feature Extraction obtains information from multi-modal inputs; Quantum Encoding maps classical features into quantum states by leveraging quantum superposition and entanglement; LQMFN achieves deep integration and dimensionality reduction of features through quantum convolution and quantum pooling; finally, the Measurement and Optimization module measures the quantum states and combines classical optimization algorithms to enhance model performance. For a detailed definition of the quantum symbols and notations used in this framework, refer to the symbol table in Appendix A.

3.1. Feature extraction

1) The input multi-modal posts for detection are represented as a set $X = \{x_1, x_2, ..., x_n\}$. Each multi-modal post $x_i \in X$, $x_i = t_i, p_i, u_i, c_i$, consists of textual content t_i, image content p_i, the author u_i, and the set of retweets/comments c_i. A multi-modal post may contain multiple retweets/comments from different users, denoted as $c_i = \{c_i^1, c_i^2, ..., c_i^j\}$. Additionally, each retweet/comment c_i^j is posted by the corresponding user z_i^j.

3.1.1. Text feature extraction

The raw text content $T = \{t_1, t_2, ..., t_n\}$ of the posts is fed into the text encoder $CLIP_{Text}$ [40] to obtain the embedded textual features f_{text}, as shown in Eq. (1).

$$f_{text}(T) = CLIP_{Text}(T), \tag{1}$$

where, $CLIP_{Text}$ adds [SOS] and [EOS] tokens to the beginning and end of the text sequence and converts it into feature vectors. To map the features into a shared multi-modal embedding space, we first normalize them and then apply a linear layer for projection.

3.1.2. Visual feature extraction

The raw image content $P = \{p_1, p_2, ..., p_n\}$ of the posts is fed into the visual encoder $CLIP_{Image}$ [40] to obtain the embedded image features f_{img}, as shown in Eq. (2).

$$f_{img}(P) = CLIP_{Image}(P), \tag{2}$$

where, $CLIP_{Image}$ is a Vision Transformer (VIT) model. $CLIP_{Image}$ divides the input image into multiple patches and converts each patch into a vector. These vectors are then processed by the Transformer layers, where the feature representations are learned through the self-attention mechanism.

3.1.3. Propagation graph feature extraction

Considering the propagation process of a post along with its publication time τ, we partition its propagation graph into a series of snapshots at fixed time intervals γ. The post itself, its comments, and the users involved in its dissemination are treated as nodes.Relationships between these nodes, such as retweets and comments, are represented as edges. Then a sequence of propagation graph snapshots is constructed at different time intervals. The propagation graph is represented as $G = \{g^1, ..., g^T\}$, where T denotes the number of time steps. Each snapshot $g^t = \{N, A^t, E^t\}$ consists of a node set N, which includes user nodes, comment nodes, and post nodes. $A^t \in \{0, 1\}^{k \times k}$ represents the adjacency matrix, indicating the relationships between nodes, including posting, commenting, and retweeting. E^t represents the set of edges. We compute the time difference between the first and last responses to a post and divide it by the number of time steps T to obtain the time interval γ, as follows:

$$\gamma = \frac{\tau_{max} - \tau_{min}}{T}, \tag{3}$$

where, τ_{min} and τ_{max} represent the time of the initial and final response to posts, respectively. Specifically, we represent the set N as a matrix $X \in \mathbb{R}^{|N| \times D}$, where D denotes the feature dimensionality. There are three types of nodes in X: sentence vectors serve as initial embeddings for posts and comments nodes, while the average embedding of posts posted by users serves as the initial embedding for the user nodes. Inspired by [41], we utilize a Dynamic Graph Network (DGN) to extract the propagation features of a post. Specifically, the structural attention layer captures the neighborhood structural information among nodes, followed by the temporal attention layer that captures the node dependencies between different time steps, learning the latent representation

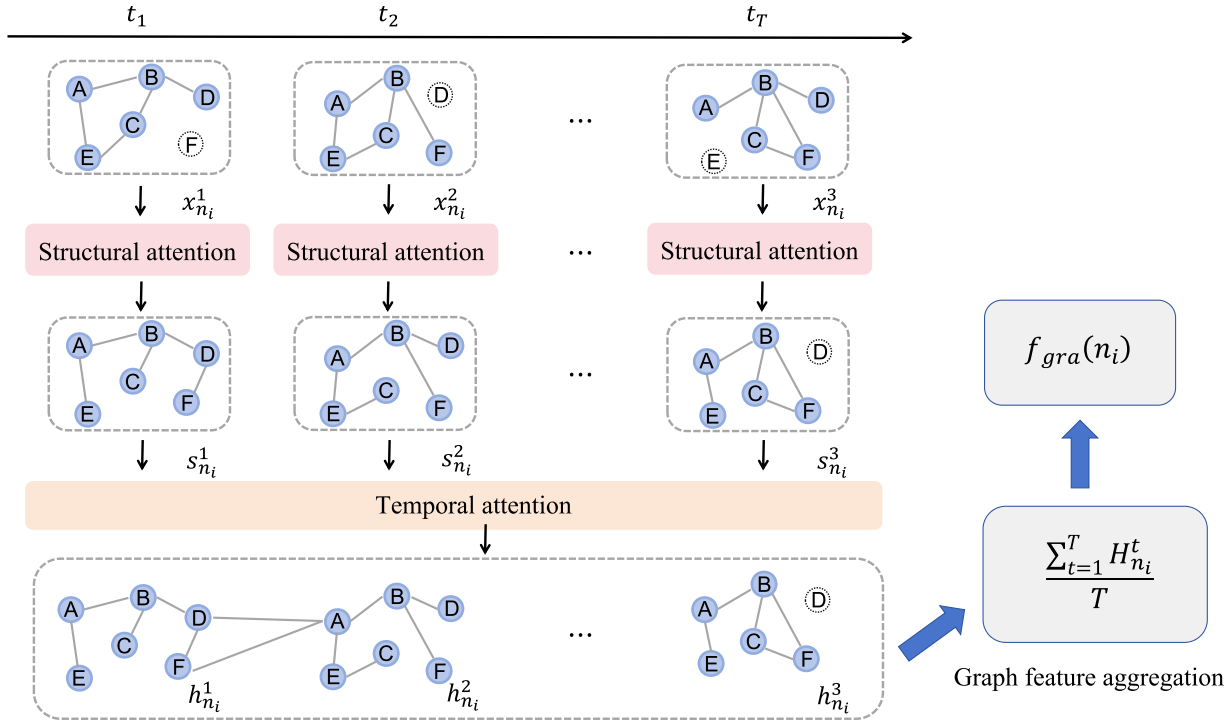


Fig. 2. Neural architecture of Dynamic Graph Network (DGN).

of nodes in the time sequence. Finally, we average the vectors of node n_i across all time steps to obtain its propagation graph embedding feature representation. The specific method is illustrated in Fig. 2.

1) Structural Attention

The structural attention layer is specifically designed to process graph data, aiming to capture the structural relationships among nodes and generate enriched node representations. For a node n_i, its initial representation $x_{n_i} \in \mathbb{R}^D, \forall n_i \in \mathcal{N}$ serves as the input to this layer, with dimension D. The output is $s_{n_i} \in \mathbb{R}^F, \forall n_i \in N$, with dimension F, representing a new representation of node n_i that integrates its local structural characteristics in the snapshot g. For node n_i, the structural attention focuses on the nodes in its neighborhood set $\mathcal{N}(n_i)$. By computing attention weights α_{ij} for node n_i, the layer selectively aggregates information from its neighbors to form an updated representation.

$$s_{n_i} = \sigma\left(\sum_{n_j \in \mathcal{N}_{(n_i)}} \alpha_{ij} W^s x_{n_j}\right), \tag{4}$$

$$\alpha_{ij} = \frac{exp(x_{ij})}{\Sigma_{n_k \in \mathcal{N}(n_i)} exp(x_{ik})}, \tag{5}$$

$$x_{ij} = LeakyReLU\left(A_{ij} \cdot a^T \left[W^s x_{n_j} || W^s x_{n_i}\right]\right) \quad \forall (i,j) \in E, \tag{6}$$

where, $\mathcal{N}(n_i)$ denotes the neighbor set of node n_i, σ is an activation function. α_{ij} denotes the attention weight capturing the interaction between n_i and its neighbor n_j, and W^s is a query matrix used to linearly transform node representations to facilitate the attention computation. A_{ij} is a trainable parameter that models the correlation between nodes n_i and n_j. $a \in \mathbb{R}^{2D}$ is a learnable vector used in computing the attention weights. || denotes concatenation, and the LeakyReLU activation function is applied to effectively model complex relationships. For node pairs with no connections, their attention weight is zero. To efficiently handle sparse graph connections, we employ sparse matrices to implement masked self-attention over neighboring nodes. Through this process, the structural attention layer can combine the local structural feature of nodes with the importance of their neighboring nodes, generating richer nodes features that better capture the structural characteristics of them in the propagation graph.

2) Temporal Attention

Temporal attention mechanism can capture nodes dependencies between different time steps in dynamic graphs and extract latent representation of nodes in time sequence. By designing a temporal attention layer, which updates the node representations in the time series, we can grasp the temporal evolution patterns of nodes in the propagation dynamic graphs. Considering the representation sequence S_{n_i} of node n_i at different time steps, we define it as $S_{n_i} = \{s_{n_i}^1, s_{n_i}^2, ..., s_{n_i}^T\}$, where $s_{n_i}^t \in \mathbb{R}^{D'}$ and $S_{n_i} \in \mathbb{R}^{T \times D'}$, D' is the dimension of inputted representations. The output of this layer is denoted as $H_{n_i} = \{h_{n_i}^1, h_{n_i}^2, ..., h_{n_i}^T\}$, where $h_{n_i}^t \in \mathbb{R}^{F'}$ and $H_{n_i} \in \mathbb{R}^{T \times F'}$. In S_{n_i}, each row denotes the inputted vector of n_i at a specific time step. In H_{n_i}, each row represents the corresponding output representation of node n_i in the layer at the respective time step.

The core goal of temporal attention is to grasp the evolving process of nodes in the propagation graph. For node n_i, the inputted representation $s_{n_i}^t$ of it contains the information of its local neighborhood structure. We use $s_{n_i}^t$ as a query to explore the historical representations of node n_i and track the dynamic changes of its neighborhood. The temporal attention can attend to the feature vectors of nodes themselves, aiding in parallel processing between nodes. By employing the Scaled Dot-Product Attention mechanism formula [42], we can determine the outputted vector of n_i at time t. The sequence S_{n_i} is projected into different spaces using linear projection matrices $W_q \in \mathbb{R}^{D' \times F'}$, $W_k \in \mathbb{R}^{D' \times F'}$, and $W_v \in \mathbb{R}^{D' \times F'}$ to obtain the queries, keys, and values, respectively. We allow the attention mechanism to consider the information from all preceding time steps up to t, to maintain the autoregressive property of the model. The temporal attention formula is expressed as:

$$H_{n_i} = \beta_{n_i}(S_{n_i} W_v), \tag{7}$$

$$\beta_{n_i}^{mv} = \frac{exp(e_{n_i})}{\Sigma_{k=1}^{T} exp\left(e_{n_i}^{mv}\right)}, \tag{8}$$

$$e_{n_i}^{mv} = \left(\frac{\left(\left(S_{n_i} W_q \right) \left(S_{n_i} W_k \right)^T \right)_{mv}}{\sqrt{F'}} + M_{mv} \right), \tag{9}$$

where, $\beta_{n_i} \in \mathbb{R}^{T \times T}$ represents the attention weight matrix derived from the multiplicative formula, $\beta_{n_i}^{mv}$ is the (m,v)-th element of the matrix β_{n_i}. F' is the dimensionality of the node n_i representation. $M \in \mathbb{R}^{T \times T}$ is the masking matrix used to enforce the autoregressive property. The definition of M is as follows:

$$M_{mv} = \begin{cases} 0, & m \le v \\ -\infty, & otherwise \end{cases}. \tag{10}$$

When $m \le v$, $M_{\mathrm{mv}} = 0$ means allowing information to flow from time step m to v. When $m > v$, $M_{\mathrm{mv}} = -\infty$, resulting in $\beta_{\mathrm{n_i}}^{\mathrm{mv}} = 0$, it means shuting off attention from time step m to v, ensuring the autoregressive property of the model.

3) Propagation Graph Embedding

To aggregating the feature vectors of n_i in the graph across all time steps and obtaining the propagation graph embedding of each node n_i, we sum the feature representations $h_{n_i}^{\mathrm{t}}$ outputted at all time steps t, and then divide it by the time steps T in graph G, yielding the propagation graph embedding $f_{gra}(n_i)$:

$$f_{gra}(\mathrm{n_i}) = \frac{\sum_{t=1}^{T} h_{n_i}^{t}}{T}, \tag{11}$$

where, $h_{n_i}^{t}$ is the feature representation of node n_i at time step t.

3.2. Quantum encoding

In quantum encoding, there are two commonly used methods: basis embedding and amplitude embedding [43]. Although basis embedding is easier to be implemented, it will demand more qubits to solve high-dimensional data. Consequently, we adopts the amplitude embedding, achieving efficient encoding of large amounts of information. In amplitude embedding, a system consisting of n qubits has 2^n probability amplitudes, capable of storing 2^n real. Considering a real vector $x = (x_1, \cdots, x_{2^n})^T$ with 2^n-dimension, the corresponding quantum state can be represented as:

$$|\psi\rangle = \frac{1}{c} \sum_{i=1}^{2^n} x_i \, |i\rangle, \tag{12}$$

where $C = \sqrt{\sum_{i=1}^{2^n} |x_i|^2}$ is a normalized constant. Considering a four-dimensional classical vector $x = (x_1, x_2, x_3, x_4)$ as an example, we use amplitude embedding to encode it into two qubits, with its original quantum state as $|00\rangle_{12}$. First, we apply a rotation gate $R_1(\theta_1)$ to the first qubit, yielding:

$$|0\rangle_1 \rightarrow \left(\sqrt{|x_1|^2 + |x_2|^2} \right) |0\rangle + \left(\sqrt{|x_3|^2 + |x_4|^2} \right) |1\rangle. \tag{13}$$

Then, a rotation gate $R_2(\theta_2)$ is applied to the second qubit:

$$|0\rangle_2 \rightarrow \frac{x_1|0\rangle + x_2|1\rangle}{\sqrt{|x_1|^2 + |x_2|^2}}. \tag{14}$$

Assuming there exists a unitary gate U such that:

$$U\left(\frac{x_1|0\rangle + x_2|1\rangle}{\sqrt{|x_1|^2 + |x_2|^2}} \right) = \frac{x_3|0\rangle + x_4|1\rangle}{\sqrt{|x_3|^2 + |x_4|^2}}. \tag{15}$$

The quantum circuit realizes a mapping from the first qubit to the second qubit through the control-gate U, where the first qubit serves as the control bit, as shown in Fig. 3.

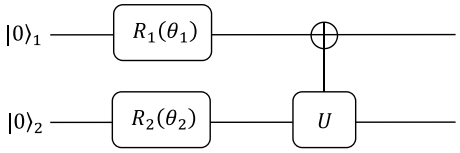


Fig. 3. Two qubits encoding.

After amplitude encoding, the quantum circuit representing x is obtained as follows:

$$\begin{aligned} |00\rangle_{12} \rightarrow & \left(\sqrt{|x_1|^2 + |x_2|^2} \right) |0\rangle \frac{x_1|0\rangle + x_2|1\rangle}{\sqrt{|x_1|^2 + |x_2|^2}} \\ & + \left(\sqrt{|x_3|^2 + |x_4|^2} \right) |1\rangle \frac{x_3|0\rangle + x_4|1\rangle}{\sqrt{|x_3|^2 + |x_4|^2}} \\ = & \, x_1|00\rangle + \, x_2|01\rangle + x_3|10\rangle + x_4|11\rangle. \end{aligned} \tag{16}$$

It is evident that only $[log_2 n]$ qubits can represent an n-dimensional vector. By concatenating features extracted from text, images, and propagation graphs, and processing them through fully connected layers the compressed multi-modal features $X = (x_1, x_2, ..., x_N)$ are obtained, where $N = 256$. Therefore, in LQMFN, we use $\left[\log_2^{256}\right] = 8$ qubits to represent the features.

3.3. Lightweight quantum-driven multi-modal fusion network

Inspired by Li et al. [35] and Qu et al. [39], we design a lightweight quantum-driven multi-modal fusion network (LQMFN) for further interaction and fusion of multi-modal features. The LQMFN consists of quantum convolutional layers and quantum pooling layers.

3.3.1. Quantum convolutional layer

The local connectivity and shared weights are prominent features in classical convolutional layers. In LQMFN, the design of quantum convolutional layers achieves effects comparable to classical convolutional layers. Specifically, quantum convolutional layers are realized through a series of two-qubit gates. These gates are similar to the classical convolutional kernels, performing entanglement operations on adjacent qubits, which reflecting the local connectivity property of LQMFN models. Additionally, all quantum gates in the quantum convolutional layer share the identical parameters, reflecting the shared weights. In this layer, the Controlled-NOT (CNOT) gate serves as the core mechanism for feature interaction. When a CNOT gate entangles a control qubit with a target qubit, the state of the target qubit becomes dependent on the state of the control qubit, enabling the exchange and fusion of local probability amplitude information between adjacent qubits. This process captures local correlations in the quantum register. The parameterized rotation gates $R_y(\theta)$ and $R_z(\theta)$ further adjust the amplitude and phase of each qubit, adapting to different feature patterns in a learnable manner. By alternating entanglement operations (CNOT gates) with parameterized rotations, the quantum convolutional layer establishes a hierarchical interaction pattern in which local features are repeatedly exchanged, transformed, and integrated. The circuit design of this layer is shown as Fig. 4, consisting of a series of quantum gates. First, a Z-axis rotation gate is applied to qubit b_1 to prepare the initial phase for subsequent operations. Then, entanglement between qubit b_1 and qubit b_0 is generated by a CNOT gate controlled by qubit b_1. Next, a Z-axis rotation gate $Rz(\theta_0)$ is applied to qubit b_0 to adjust the phase of qubit b_0. Subsequently, a Y-axis rotation gate $Ry(\theta_1)$ is applied to qubit b_1 to adjust the amplitude and phase of qubit b_1. After that, entanglement and information transfer between qubit b_0 and qubit b_1 are further performed by the second CNOT gate. Then, a Y-axis rotation gate $R_y(\theta_2)$ is applied to qubit b_1 to further adjust the state of qubit b_1. Finally, entanglement is established again through the third CNOT gate, and a Z-axis rotation gate is applied to qubit b_0 to make final adjustments to the phase of qubit

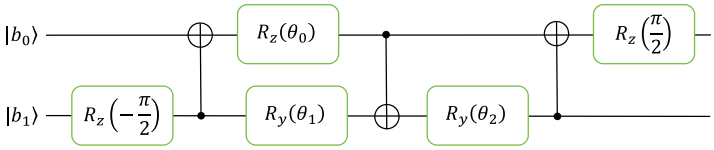


Fig. 4. The circuit of quantum convolutional layer.

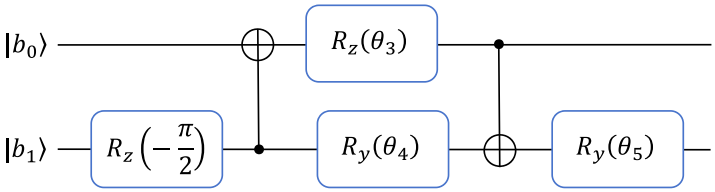


Fig. 5. The circuit of quantum pooling layer.

b_0. Through the quantum convolutional layers, the initial quantum state $|\psi\rangle$ is transformed into:

$$|\psi_{\text{conv}}\rangle = U_{\text{conv}}^{(0,1)}(\theta) \cdot |\psi\rangle, \tag{17}$$

$$\begin{aligned} U_{\text{conv}}^{(0,1)}(\theta) = {} & R_z\left(\frac{\pi}{2}\right)_{b_0} \cdot CNOT_{b_1 \to b_0} \cdot R_y(\theta_2)_{b_1} \cdot CNOT_{b_0 \to b_1} \\ & \cdot R_y(\theta_1)_{b_1} \cdot R_z(\theta_0)_{b_0} \cdot CNOT_{b_1 \to b_0} \cdot R_z\left(-\frac{\pi}{2}\right)_{b_1}. \end{aligned} \tag{18}$$

where $U_{\text{conv}}^{(0,1)}(\theta)$ represents the convolution operation on qubits $|b_0\rangle$ and $|b_1\rangle$ and $CNOT_{b_1 \to b_0}$ indicates that qubit $|b_1\rangle$ controls qubit $|b_0\rangle$. The matrix representations of the quantum gates are as follows:

$$R_y(\theta) = \begin{pmatrix} cos\frac{\theta}{2} & -sin\frac{\theta}{2} \\ -sin\frac{\theta}{2} & cos\frac{\theta}{2} \end{pmatrix}, R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}, \tag{19}$$

where, θ is a trainable parameter that can be updated during training to adapt to various operations of the quantum circuit.

3.3.2. Quantum pooling layer

Analogous to pooling operations in classical convolutional neural networks, which reduce the resolution of feature maps while preserving key information, the quantum pooling layer compresses the information of multiple qubits into fewer qubits, thereby reducing the dimensionality of the quantum state. This process is based on variational quantum gates, combining entanglement and learnable rotations to achieve feature fusion and compression.

In the pooling operation, CNOT gates remain the primary tool for feature interaction, allowing probability amplitude information from one qubit to be transferred and fused into another, thus constructing a shared quantum state space. The parameterized rotation gates $R_y(\theta)$ and $R_z(\theta)$ are then applied to adjust the amplitude and phase distribution of the compressed qubits, ensuring that the reduced representation retains the most salient information from the original multi-qubit state. This controlled fusion and compression mechanism is analogous to classical pooling, where local neighborhood information is aggregated into a single representative value. The circuit of the quantum pooling layer based on variational quantum gates is shown as Fig. 5. First, a Z-axis rotation gate is applied to qubit b_1 to prepare the initial phase. Then, entanglement between the two qubits is generated through the CNOT gate. Next, a Z-axis rotation gate $R_Z(\theta_3)$ is applied to qubit b_0 to adjust the phase of qubit b_0, and a Y-axis rotation gate $R_y(\theta_4)$ is applied to qubit b_1 to adjust the amplitude and phase of qubit b_1. Finally, entanglement is further established between qubit b_0 and qubit b_1 through the second CNOT gate, and a Y-axis rotation gate $R_y(\theta_5)$ is applied to qubit b_1 to adjust the amplitude and phase of qubit b_1 again. The pooled quantum state is obtained, where θ is the trainable parameter.

Through quantum pooling layers, the convoluted quantum state $|\psi_{\text{conv}}\rangle$ is transformed into:

$$|\psi_{poll}\rangle = U_{poll}^{(0,1)}(\theta) \cdot |\psi_{conv}\rangle, \tag{20}$$

$$U_{poll}^{(0,1)}(\theta) = R_y(\theta_5)_{b_1} \cdot \text{CNOT}_{b_0 \to b_1} \cdot R_y(\theta_4)_{b_1} \cdot R_z(\theta_3)_{b_0} \cdot \text{CNOT}_{b_1 \to b_0} \cdot R_z\left(-\frac{\pi}{2}\right)_{b_1}, \tag{21}$$

where $U_{poll}^{(0,1)}(\theta)$ represents the pooling operation on qubits $|b_0\rangle$ and $|b_1\rangle$. The variational quantum gates are applied to adjacent qubits, achieving feature fusion of local probability amplitudes while performing dimensionality reduction and sampling.

3.3.3. LQMFN quantum circuit

We construct a three-layer LQMFN circuit, shown as Fig. 6. Let the input qubits be n, and in the first layer, quantum convolution and pooling are performed on $\frac{N}{2}$ pairs of adjacent qubits, respectively.

$$|\psi_1\rangle = \prod_{k=0}^{\frac{N}{2}-1} \left(U_{pool}^{(2k,2k+1)} U_{conv}^{(2k,2k+1)}\right) |\psi\rangle, \tag{22}$$

where $U_{\text{conv}}^{(2k,2k+1)}$ and $U_{\text{pool}}^{(2k,2k+1)}$ represent the convolution and pooling operations on the k-th pair of qubits $(2k, 2k+1)$, respectively. In the second layer, quantum convolution and pooling operations are performed on $\frac{N}{4}$ pairs of qubits:

$$|\psi_2\rangle = \prod_{k=0}^{\frac{N}{4}-1} \left(U_{pool}^{(2k+1,2k+3)} U_{conv}^{(2k+1,2k+3)}\right) |\psi_1\rangle. \tag{23}$$

In the third layer, quantum convolution and pooling operations are performed on $\frac{N}{8}$ pairs of qubits

$$|\psi_3\rangle = \prod_{k=0}^{\frac{N}{8}-1} \left(U_{pool}^{(2k+3,2k+7)} U_{conv}^{(2k+3,2k+7)}\right) |\psi_2\rangle. \tag{24}$$

To optimize resource usage and reduce computational costs, some qubits in the circuit are discarded. These discarded qubits cease to participate in subsequent quantum operations, effectively reducing data dimensions and computational complexity while retaining key feature information. Through the three layers of quantum convolution and pooling, the overall operation of the LQMFN can be expressed as:

$$\begin{aligned} |\psi'\rangle = {} & \prod_{k=0}^{\frac{N}{8}-1} \left(U_{\text{pool}}^{(2k+3,2k+7)} U_{\text{conv}}^{(2k+3,2k+7)}\right) \prod_{k=0}^{\frac{N}{4}-1} \left(U_{\text{pool}}^{(2k+1,2k+3)} U_{\text{conv}}^{(2k+1,2k+3)}\right) \\ & \times \prod_{k=0}^{\frac{N}{2}-1} \left(U_{\text{pool}}^{(2k,2k+1)} U_{\text{conv}}^{(2k,2k+1)}\right) |\psi\rangle. \end{aligned} \tag{25}$$

Finally, a quantum measurement is performed on qubit b_7 to obtain the measurement result. LQMFN fully leverages the quantum gates to

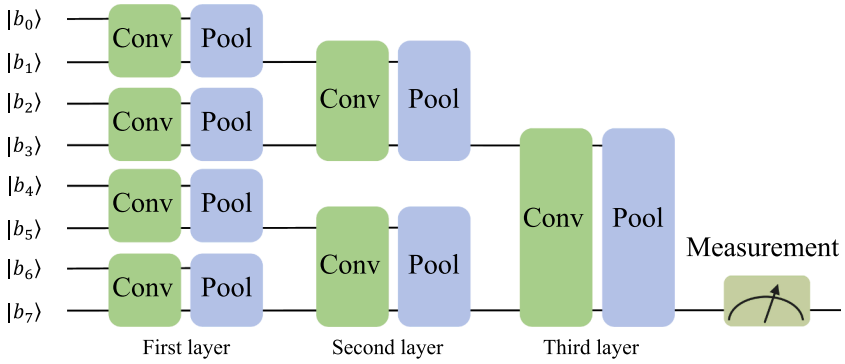


Fig. 6. The LQMFN circuit in LQMF-RD.

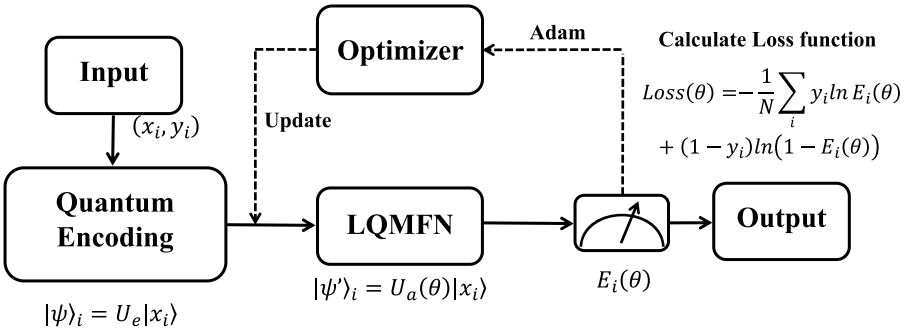


Fig. 7. Measurement and Optimization.

give the circuit strong information expression and entanglement capabilities, allowing it to extract discriminative features that are helpful for rumor detection. This helps to more effectively integrate the correlations between features, thereby enabling LQMF-RD to achieve better detection performance (Fig. 7).

3.4. Qubits measurement and optimization

To promote the performance of LQMF-RD, we obtain the expected values of qubits through multiple measurements to predict the model's output. The cross-entropy loss function is used for evaluating the results of LQMF-RD. We use the Adam optimizer for parameters optimization, which utilizes the gradient descent algorithm and automatically adjusts the learning rate. The specific steps of the measurement and optimization process are as follows:

Set a classical input data (x_i, y_i), where x_i represents the i-th input data containing information from three modalities, and y_i represents the corresponding label. Real information labels are set to 1, and rumor labels are set to 0. The original quantum state is set to $|0\rangle^{\otimes n}$. After the unitary transformation U_e in the encoding layer, x_i is transformed into the quantum state $|\psi\rangle_i = U_e|x\rangle$. The unitary transformation $U_a(\theta)$ of the LQMFN layer consists of a series of parameterized quantum gates $U_k(\theta_k)$, where θ is the trainable parameter. After applying $U_a(\theta)$,the quantum state is updated to $|\psi'\rangle_i = U_a(\theta)|\psi\rangle_i$. Using the measurement operator $M = I^{\otimes(n-1)} \otimes Z$, the system is measured to obtain the expected value of the prediction:

$$\begin{aligned} E_i(\theta) &= \langle\psi'|M|\psi'\rangle_i \\ &= \langle\psi|U_a^\dagger(\theta) M U_a(\theta)|\psi\rangle_i. \end{aligned} \tag{26}$$

Define cross-entropy as the loss function:

$$Loss(\theta) = -\frac{1}{N}\sum_i y_i\, lnE_i(\theta) + (1-y_i) ln\big(1-E_i(\theta)\big). \tag{27}$$

Calculate the partial derivatives of E relative to θ_k using the parameter-shift rule:

$$\frac{\partial E(\theta)}{\partial \theta_k} = \left\langle \psi \middle| U_a^\dagger M \frac{\partial U_a}{\partial \theta_k} \middle| \psi \right\rangle + H.c, \tag{28}$$

where, H.c is the complex conjugate of $\left\langle \psi \middle| U_\alpha^\dagger M \frac{\partial U_\alpha}{\partial \theta_k} \middle| \psi \right\rangle$. In this circuit, all quantum gates are Pauli rotation gates. According to [44], the gradient calculation can be simplified to:

$$\frac{1}{2}\langle\langle\psi|U_a^\dagger\Big(\theta_k+\frac{\pi}{2}\Big) M U_a\Big(\theta_k+\frac{\pi}{2}\Big)|\psi\rangle - \left\langle\psi\middle|U_a^\dagger\Big(\theta_k-\frac{\pi}{2}\Big) M U_a\Big(\theta_k-\frac{\pi}{2}\Big)\middle|\psi\right\rangle. \tag{29}$$

By updating parameters in this manner, LQMFN can be effectively optimized to perform more accurate detection tasks.

4. Experiments and results

4.1. Dataset and experimental setup

4.1.1. Datasets

We utilize two publicly available rumor detection datasets, Weibo [45] and PHEME [46], to validate the proposed model. The Weibo is a Chinese rumor dataset collected from the Sina Weibo platform, while PHEME includes tweets related to several different breaking news on Twitter. Both datasets include text, images, and comments for each tweet. To ensure a fair comparison of multi-modal false information detection methods, we remove any entries with missing modalities or duplicate data from the datasets. The processed datasets are detailed in Table 1.

4.1.2. Experimental setup and evaluation metrics

We implement the model proposed using the PyTorch and PennyLane frameworks. PennyLane is an open-source library specifically designed for quantum machine learning, enabling the efficient implementation of QNN and hybrid quantum-classical algorithms. The textual

Table 1
Dataset statistics.

Dataset	Fake Information	Non-Fake Information	Number of Images	Number of Users	Number of Comments	Number of Nodes in Propagation Network
Weibo	590	877	1467	985	4564	6963
PHEME	590	1428	2018	894	7388	4329

Table 2
Training setting details.

Parameter	Weibo	PHEME
learning rate	1e-4	1e-4
batch size	16	16
epoch	20	20
max length	50	50
weight decay	5e-4	5e-4
dropout	0.6	0.5
qubits	8	8
time steps	3	3

and visual features are extracted using CLIP models respectively to obtain high-quality multi-modal feature representations. To address the language differences between Chinese texts (Weibo) and English texts (PHEME), we employ specialized text encoders for feature extraction. Specifically, Chinese texts are encoded using the Chinese-CLIP encoder, while English texts use the original CLIP model. Both encoders map the text into a unified multi-modal embedding space, ensuring consistency and feasibility of fusion across different languages.

During data preprocessing, we randomly divide the dataset into training, validation, and test sets with a 7:1:2 ratio. The experiments are conducted on dual NVIDIA RTX 3090 GPUs (24G). During training, we use 8 qubits, and choose the time steps from {1, 2, 3, 4, 5}. We use cross-entropy loss function and Adam optimizer for parameters optimization. The specific hyperparameter settings are listed in Table 2. To comprehensively evaluate the performance of LQMF-RD, we adopt accuracy, precision, recall, and F1 score as evaluation metrics.

4.2. Baseline models and performance comparison

4.2.1. Baseline models

To evaluate the effectiveness of LQMF-RD, we compare it with several classical models that leverage textual features, visual features, and propagation structure features of posts.

- QSAN [47]: This model integrates quantum-driven text encoding and symbolic attention mechanisms to extract features of rumors.
- SAFE [48]: This model utilizes TextCNN to grasp textual and image characteristics in fake news and introduces fine-tuned cosine similarity across these two textual modalities into the loss function.
- MVAE [10]: This model employs a bimodal variational autoencoder to concatenate the text and visual features, and reconstructs the two modalities using decoders.
- EANN [11]: This model captures text and image features by TextCNN and VGG-19 respectively, and then combines these modalities for fake news detection.
- GLAN [49]: The model integrates a heterogeneous graph framework that combines local semantics and global structural information. It is employed to effectively detect rumors by simultaneously analyzing tweet content and its propagation patterns.
- Bi-GCN [8]: This model employs a bidirectional GCN for capturing propagation characteristics of the rumor based on its two propagation directions.
- MFAN [50]: This model captures text, visual, and propagation graph characteristics of rumor, and uses cross-modal attention mechanisms to improve the ability of features extraction across modalities.

Table 3
Main results on Weibo.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
QSAN	71.01	71.02	67.54	67.58
SAFE	84.95	84.98	84.95	84.96
EANN	83.22	82.56	82.14	82.27
MVAE	71.76	70.52	70.21	70.32
GLAN	84.46	84.71	82.57	83.19
Bi-GCN	86.72	86.63	86.61	86.62
MFAN	88.95	88.91	88.13	88.33
MVFN	89.36	89.20	88.69	88.95
TD-MMC	90.17	89.68	89.76	89.72
GMCL	91.59	91.30	90.52	90.91
LQMF-RD	**92.20**	**92.22**	**92.20**	**92.21**

- MVFN [51]: This model achieves an effective multi-modal fake news detection method by utilizing graph attention mechanisms and modality alignment strategies to integrate textual, visual, and social context information in fake news.
- TD-MMC [52]: This model encodes and fuses textual, image, and social network features of fake news, and uses cross-modal attention and cosine similarity to capture the complementarity and inconsistencies among these features.
- GMCL [53]: This model effectively integrates textual, image, and social graph features through semantic correlation-assisted cross-modal contrastive learning.

Among these, QSAN only uses text data for rumor detection, while SAFE uses both text and image data. GLAN and Bi-GCN use propagation features from tweets for rumor detection, and MFAN, MVFN, TD-MMC, and GMCL detect rumors based on multi-modal data.

4.2.2. Comparative analysis of experimental results

Tables 3 and 4 present the experimental results of LQMF-RD compared with baseline methods on the Weibo and PHEME datasets. Boldface indicates the optimal performance in the tables. The results demonstrate that LQMF-RD surpasses all baseline methods across all evaluation metrics for rumor detection on both datasets.

First, the results show that graph topology-based rumor detection methods (GLAN, Bi-GCN, MFAN, MVFN, TD-MMC, GMCL, and LQMF-RD) generally outperform traditional deep learning methods (QSAN, SAFE, EANN, MVAE). This is because these graph topology-based methods can effectively utilize the structural information in the rumor dissemination graph. The connections between nodes significantly impact information dissemination. Graph topology-based methods can better capture these connection relationships, thereby more accurately identifying patterns of rumor propagation.

Secondly, multi-modal fusion methods (MFAN MVFN, TD-MMC, GMCL, and LQMF-RD) significantly outperform the methods that only consider propagation features (GLAN, BiGCN). This is because multi-modal methods can not only effectively capture the propagation characteristics of posts in various directions from a global perspective, but also combine the textual and image data of them, thereby enhancing the model's detection performance.

Compared to MFAN, MVFN, TD-MMC, and GMCL, LQMF-RD achieves an average accuracy improvement of 2.18% and 1.15% on the two datasets, respectively. The enhancement is partly due to structural and temporal attention in LQMF-RD to extract features of post

Table 4
Main results on PHEME.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
QSAN	75.13	69.97	65.80	66.87
SAFE	82.33	81.10	80.31	80.38
EANN	78.09	72.46	72.26	72.13
MVAE	75.62	73.49	72.25	70.34
GLAN	83.32	81.25	77.13	79.14
Bi-GCN	84.71	84.03	83.43	83.73
MFAN	88.73	87.07	85.61	86.16
MVFN	88.71	88.05	87.67	87.86
TD-MMC	89.61	90.10	84.37	87.24
GMCL	90.15	88.39	87.61	88.01
LQMF-RD	**90.45**	**90.41**	**90.45**	**90.43**

Table 5
Model performance on Weibo and PHEME datasets.

DataSet	Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Weibo	TI	88.14	88.77	88.14	88.46
	TG	88.81	88.75	88.81	88.78
	IG	87.19	87.37	87.19	87.28
	TIG	**92.20**	**92.22**	**92.20**	**92.21**
PHEME	TI	87.44	88.02	87.44	87.73
	TG	87.94	87.83	87.94	87.89
	IG	85.93	85.76	85.93	85.85
	TIG	**90.45**	**90.41**	**90.45**	**90.43**

propagation. Structural attention enables the model to better explore relationships between nodes in the propagation graph. Temporal attention helps the model to explore dependencies between nodes at different time steps, better understanding the dynamic evolution of rumor propagation. Additionally, LQMF-RD effectively mines complex relationships between multi-modal features using LQMFN and achieves efficient fusion of text, image, and post propagation structure features.

4.3. Ablation experiments

4.3.1. Impact of different modal data on model performance

We compare the influence of single modality data on LQMF-RD. The results are as Table 5, where T represents the text modality, I represents the image modality, and G represents the propagation graph modality. On both datasets, LQMF-RD achieves the best performance under the TIG combination, with accuracies of 92.20% and 90.45%, respectively. The next best is the TG combination, with accuracies of 88.81% and 87.94%, followed by the TI combination, with accuracies of 88.14% and 87.44%. The IG combination performs the worst, with accuracies of 87.19% and 85.93%.

First, removing the propagation graph modality results in a decrease of 4.06% and 3.01% in model accuracy on the both datasets, respectively. This is primarily because the propagation graph provides crucial information about user relationships and information propagation paths, helping the model better understand the propagation patterns and impacts of false information. Removing the image modality results in a decrease of 3.39% and 2.51% in the accuracy, respectively, since image information can help the model to identify misleading or inconsistent content within the rumors. Removing the text modality results in a decrease of 5.01% and 4.52% in the accuracy, respectively, as text provides direct and essential clues about the rumor content and viewpoints. While images can deliver intuitive image information, key details in rumors–such as event time, location, and participants–are often difficult to extract accurately from images alone.

Overall, text contributes the most to fake news detection, while the dynamic propagation graph ranks second due to its integration of textual information. In contrast, the role of images is relatively small.

Table 6
DGN mechanism analysis.

DataSet	Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Weibo	None	88.81	88.76	88.81	88.79
	Structural	89.73	90.29	89.73	90.01
	Temporal	90.17	90.40	90.17	90.29
	DGN	**92.20**	**92.22**	**92.20**	**92.21**
PHEME	None	88.44	88.26	88.44	88.35
	Structural	88.94	88.85	88.95	88.90
	Temporal	89.95	89.82	89.95	89.89
	DGN	**90.45**	**90.41**	**90.45**	**90.43**

4.3.2. Attention mechanism analysis

To verify the efficacy of the Dynamic Graph Network (DGN), we perform ablation studies by removing the structural, temporal attention modules and the entire DGN architecture, respectively. The findings are in Table 6.

The results indicate that on both datasets, the performance of DGN is superior to using only structural attention or temporal attention alone, demonstrating that combining node neighborhood structures and dynamic information of nodes across different time steps can enable the model to extract more features from it and improve its performance. Compared to the None, Structural achieves reasonable results, improving accuracy by 0.92% and 0.50% on both datasets, respectively, while Temporal improves accuracy by 1.36% and 1.51%. Removing structural attention from DGN results in a decrease by 2.03% and 0.50% in the accuracy, respectively, while removing temporal attention decreases the accuracy by 2.47% and 1.51%. These results demonstrate that applying structural attention to node feature extraction within specific snapshots and temporal attention to capture dynamic features across time steps are effective. Additionally, compared to TI without propagation graph modality in Table 5, the models in Table 6 exhibit better performance, further validating the feature expression ability of the propagation graph for user relationships and information propagation paths.

4.3.3. Analysis of time steps

We investigate the efficacy of LQMF-RD in learning temporal features of nodes with different time step settings. As shown in Fig. 8, findings indicate that different time steps significantly affect the model performance. Specifically, when time steps are set to 1, the propagation graph structure is sparse, resulting in relatively low accuracy on the two datasets, 90.17% and 87.94%, respectively. With increasing time steps, the model's accuracy gradually improves, reaching its highest values of 92.20% and 90.45% on the two datasets when time steps are set to 3. However, further increasing time steps leads to a decline in accuracy. Although larger time steps can capture more subtle changes in the propagation graph, if the time step is too large, the model may excessively focus on short-term dynamic changes due to frequent updates, thereby neglecting overall trends. Therefore, we choose a time step of 3 as the optimal parameter, as it strikes a balance between sensitivity to dynamic evolution and effective capture of temporal features in the propagation graph.

4.3.4. Impact of the number of Qubits

With limited resources, we compare the performance of LQMF-RD under quantum encoding of different bits such as 4, 6, 8, and 9 qubits.

As shown in Fig. 9, the quantum encoding with 8 qubits shows the optimal results on both datasets. Relative to 4 qubits, 8 qubits improve the accuracy by 0.67% and 1% on the two datasets, respectively, indicating that 8 qubits offer an optimal balance of performance given the current resource constraints. When increasing qubits, the performance does not increase monotonically. This is associated with the difficulty of entanglement and error correction, and model capacity. Compared to 4 qubits, the 6 qubits configuration enhances feature representation ability. However, the accumulation of quantum noise affects the

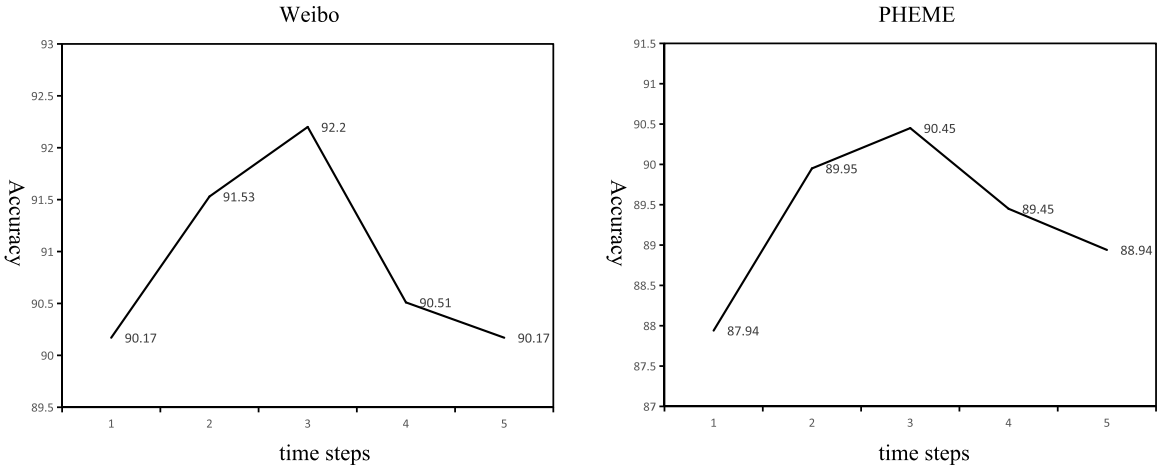


Fig. 8. The impact of different time steps on LQMF-RD.

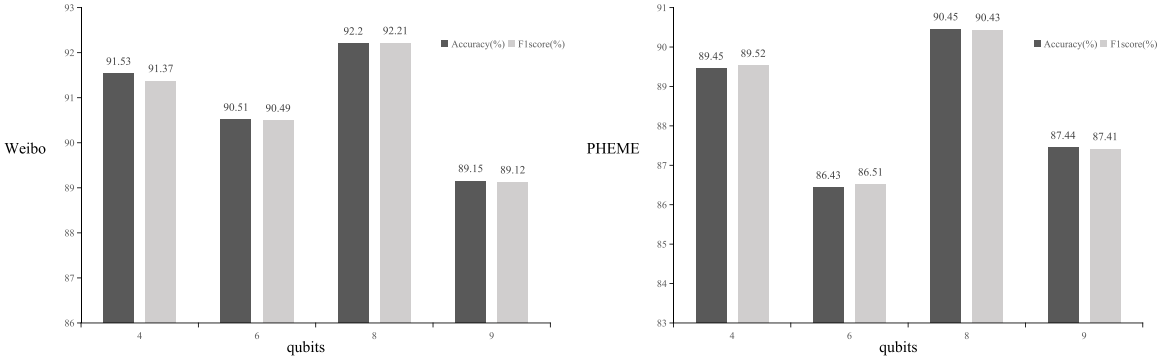


Fig. 9. The impact of different qubits on LQMF-RD.

stability of information representation in quantum state space, leading to a slight performance decline. 8 qubits provide a sufficient feature dimension, moderate noise impact, and optimal optimization efficiency. In quantum computing, the barren plateau phenomenon refers that the rapid increase in quantum state space with the number of qubits, while useful quantum states (those providing computational advantages) become sparse, while noise accumulation becomes more significant. The performance degradation of the model under quantum encoding with 9 qubits is related to this phenomenon. In conclusion, 8 qubits is the best choice for LQMF-RD.

4.4. Hyperparameter sensitivity analysis

Hyperparameters play a crucial role in machine learning models, and their proper selection directly affects the final performance of a model. Ideally, a model exhibits strong robustness, meaning that changes in hyperparameters do not lead to significant performance fluctuations [54]. To optimize the performance of the LQMF-RD model, we adopt a grid search method to tune the learning rate and dropout rate in order to determine their optimal combination. To ensure a reasonable selection of hyperparameters, refering to existing studies [8,55] and based on experimental experience, we set the candidate range for the learning rate as [$1e^{-2}$, $1e^{-3}$, $1e^{-4}$, $1e^{-5}$, $1e^{-6}$, $1e^{-7}$] and for the dropout rate as [0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8].

Through grid search, we traverse all possible hyperparameter combinations and evaluate the model performance on the validation set. The experimental results, as shown in Fig. 10, indicate that LQMF-RD maintains relatively stable performance under different hyperparameter settings, demonstrating good robustness. This shows that the model can sustain high performance over a wide range of hyperparameters and is suitable for various scenarios. However, when the learning rate is too high, the model tends to oscillate and struggles to converge; conversely, a learning rate that is too low may prolong training time and lead the model to fall into local optima. Based on our experimental analysis, we finally determine that the optimal hyperparameter combination is a learning rate of $1e^{-4}$ and a dropout rate of 0.6 for the Weibo dataset, and a learning rate of $1e^{-4}$ and a dropout rate of 0.5 for the PHEME dataset, under which the model converges quickly and achieves high performance.

4.5. Validation on cross-platform news datasets

To further evaluate the generalization capability of the proposed LQMF-RD framework across different data domains, we conducted additional experiments on the FakeNewsNet dataset [56]. Unlike typical social media rumor datasets, FakeNewsNet contains news content and associated social context information collected from fact-checking websites. The dataset is divided into two subsets: PolitiFact, which focuses on political news verification, and GossipCop, which targets entertainment news detection. Table 7 presents the comparison between LQMF-RD and several state-of-the-art methods, including MFAN [57], KAHAN [58], and DGM [59].

Compared with these models, LQMF-RD achieves superior performance across multiple evaluation metrics, indicating that the proposed quantum graph fusion mechanism has a significant advantage in extracting highly discriminative cross-modal representations. The experimental results further show that LQMF-RD maintains strong performance on

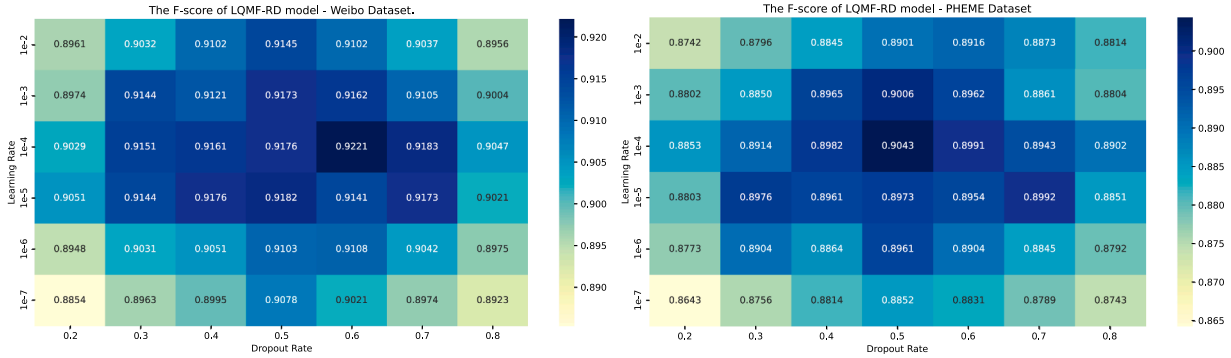


Fig. 10. Results of hyperparameter sensitivity analysis.

Table 7
Results in FakeNewsNet. Bold and underlined denote the best and second-best results, respectively.

Dataset	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
PolitiFact	MFAN	78.26	78.55	78.26	76.74
	KAHAN	<u>86.90</u>	<u>88.30</u>	<u>91.20</u>	<u>89.70</u>
	DGM	86.36	88.02	82.59	84.29
	LQMF-RD	**90.09**	**91.93**	**91.84**	**92.02**
GossipCop	MFAN	74.21	74.24	74.21	74.20
	KAHAN	<u>82.70</u>	<u>86.40</u>	<u>91.70</u>	<u>89.00</u>
	DGM	82.11	83.04	82.11	81.97
	LQMF-RD	**85.41**	**87.04**	**92.66**	**89.87**

datasets from different platforms and domains, thereby demonstrating its robust generalization capability.

4.6. Analysis of noise interference

The interference of quantum noise must be taken into account when developing Quantum Machine Learning (QML). Noise in quantum channels can potentially disrupt quantum entangled states, thereby reducing the model's accuracy. We describe single-qubit quantum noise using Kraus matrices as follows.

Bit-flip (BF) noise refers to the process in which a qubit state is flipped from $|0\rangle$ to $|1\rangle$ or vice versa with a certain probability. Its Kraus matrices are:

$$K_{BF0} = \sqrt{1-p_{BF}}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_{BF1} = \sqrt{p_{BF}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \tag{30}$$

where p_{BF} denotes the probability of a bit-flip occurring. K_{BF0} corresponds to no bit-flip (with probability $1-p_{BF}$), and K_{BF1} corresponds to a bit-flip event (with probability p_{BF}).

Phase-flip (PF) noise describes the process where the phase of the qubit flips, effectively transforming the $|+\rangle$ state into $|-\rangle$ and vice versa. Its Kraus matrices are:

$$K_{PF0} = \sqrt{1-p_{PF}}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_{PF1} = \sqrt{p_{PF}}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \tag{31}$$

where p_{PF} is the probability of a phase-flip. K_{PF0} represents no phase-flip (probability $1-p_{PF}$), and K_{PF1} represents a phase-flip event (probability p_{PF}).

Amplitude damping (AD) noise models energy dissipation in the quantum system, typically representing the decay from an excited state to the ground state. Its Kraus matrices are:

$$K_{AD0} = \begin{bmatrix}1 & 0\\ 0 & \sqrt{1-p_{AD}}\end{bmatrix}, \quad K_{AD1} = \begin{bmatrix}0 & \sqrt{p_{AD}}\\ 0 & 0\end{bmatrix}, \tag{32}$$

Table 8
The fidelity under different noise conditions.

Noise type	p = 0.01	p = 0.1
BF	97.06	70.31
PF	99.17	90.03
AD	99.99	99.98
DN	98.14	98.02
ALL	93.91	60.07

where p_{AD} is the probability of energy dissipation. K_{AD0} corresponds to no energy loss (probability $1-p_{AD}$), and K_{AD1} corresponds to an energy decay event (probability p_{AD}).

Depolarizing (DN) noise represents the process by which the qubit becomes completely mixed, losing all information about its original state. Its Kraus matrices are:

$$\begin{aligned} K_{DN0} &= \sqrt{1-p_{DN}}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_{DN1} = \sqrt{\frac{p_{DN}}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix},\\ K_{DN2} &= \sqrt{\frac{p_{DN}}{3}}\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}, \quad K_{DN3} = \sqrt{\frac{p_{DN}}{3}}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \end{aligned} \tag{33}$$

where p_{DN} is the depolarizing noise intensity. K_{DN0} denotes no depolarization (probability $1-p_{DN}$), while K_{DN1}, K_{DN2}, and K_{DN3} correspond to depolarization effects associated with different Pauli operators, each occurring with probability $p_{DN}/3$.

The performance of LQMFN to noise can be measured by fidelity, which is an indicator of the similarity among two quantum states. Its expression is:

$$F(\rho,\sigma) = \left(Tr\left[\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right]\right)^2, \tag{34}$$

where ρ and σ are the density matrices of the two mixed states. The fidelity of LQMFN under four individual noise channels and a composite one containing all kinds of noise is calculated, with noise probabilities set at 0.1 and 0.01. "ALL" indicatess superposition of these four different noises on each qubit. The experimental findings are as Table 8. The experimental results indicate that LQMFN exhibits high robustness to quantum noises and BF noise has the most significant impact on the model's fidelity. BF noise causes direct flipping of the qubit state leading to the loss of encoded feature information. Compared to BF noise, PF and AD noise have less impact on LQMFN fidelity. These two kinds of noises mainly constrain the localization of quantum circuit without causing direct information lost. Therefore, even in the presence of these noises, LQMFN can maintain high fidelity. Additionally, with increasing noises of BF, its impact becomes ever more significant, because errors caused by qubit flips propagate through multiple layers of the circuit, where quantum entanglement amplifies their effect, resulting in a greater overall impact on fidelity. If a qubit flips,

Table 9
The comparison of time complexity.

Model	Time Complexity
SAFE	$O(n)$
EANN	$O(n)$
MVAE	$O(n)$
GLAN	$O(n)$
Bi-GCN	$O(n)$
MFAN	$O(n)$
LQMF-RD	$O(log_2 n)$

this error will be amplified through quantum entanglement in subsequent layers, leading to cumulative errors. Despite the significant impact of BF noise on LQMFN, the experimental results show that LQMFN maintains high fidelity even at a noise level of $p = 0.1$, demonstrating the robustness of LQMFN against a certain degree of quantum noise interference. For PF and AD noise, the fidelity is nearly 1, indicating that these kinds of noises have a minimal impact on LQMFN performance.

4.7. Complexity analysis

4.7.1. Quantum circuit complexity

We evaluate the complexity of quantum circuit by calculating the quantity of quantum gates in it. In the quantum convolutional layer, we used 8 quantum gates, and in the quantum pooling layer, we used 6 quantum gates. For a system composed of N qubits ($N = 2^a, a \geq 1$ and is an integer), the ith layer requires $14N/2^i$ quantum gates, where $i \leq a$. Therefore, the calculation of the required quantity of quantum gates for LQMFN is as follows:

$$\begin{aligned} &2 \cdot \left(\frac{N}{2} \cdot 7 + \frac{N}{2^2} \cdot 7 + \cdots + \frac{N}{2^{a-1}} \cdot 7 \right) + 14 \\ &= 7N\left(2 - \frac{1}{2^{a-2}}\right) + 14 = 14(N-1). \end{aligned} \tag{35}$$

Thus the complexity of LQMFN circuit is $O(n)$.

4.7.2. Time complexity

In deep learning models, lower time complexity usually suggests higher computing efficiency of the model. Comparing LQMFN in LQMF-RD with traditional neural networks in baselines, as shown in Table 9, for traditional neural networks, the time complexity of generating n-dimensional classical data is $O(n)$, meaning that the computational resources required increase linearly with the data dimension. However, LQMFN leverages the advantage of quantum amplitude encoding, requiring only $\log_2 n$ qubits to symbolize a vector with n-dimension, which significantly reduces the circuit's time complexity to $O(\log_2 n)$. This efficient representation capability of qubits allows LQMFN to capture more information with fewer resource consumption.

4.8. Effectiveness analysis

To validate the effectiveness of LQMFN in multimodal rumor detection tasks, we conduct comparative experiments by replacing the LQMFN layer with two mainstream feature fusion methods: Cross Attention and Concatenation (Concat). Cross Attention achieves information interaction by computing cross-modal attention weights, while Concat directly concatenates different modal features and learns through fully connected layers.

The experimental results in Table 10 show that LQMFN achieves superior performance in both accuracy (92.20%) and F1 score (92.21%), outperforming the Cross Attention and Concat methods. This indicates that LQMFN has significant advantages in integrating multimodal information, effectively enhancing rumor detection capabilities. Additionally, LQMFN requires updating only 0.01M training parameters, while Cross Attention and Concat require 0.93M and 0.79M parameters, respectively. This demonstrates that LQMFN, while maintaining high performance, greatly reduces computational complexity and storage requirements. Therefore, LQMFN not only outperforms traditional methods in terms of performance but also provides significant advantages in terms of computational efficiency and resource consumption due to its lightweight design, making it highly suitable for resource-constrained environments.

Table 10
Performance and Efficiency Comparison of Different Fusion Methods on Weibo Dataset (Params indicates the adjustable parameters during training of the fusion network.).

Method	Accuracy(%)	F1 score(%)	Params(M)
Cross Attention	91.14	91.07	0.93M
Concat	89.31	88.76	0.79M
LQMFN	92.20	92.21	0.01M

5. Conclusion

In this paper, we propose LQMF-RD, a lightweight quantum-driven multi-modal feature fusion framework for social media rumor detection. LQMF-RD utilizes the Dynamic Graph Network (DGN) to analyze the structure and dynamic evolution of rumor propagation graph, using structural and temporal attention mechanisms to capture the node neighborhood features and temporal features in the graph. This enables the model to comprehensively understand the propagation patterns of rumors and provide accurate detection results at different time stages. Furthermore, the introduction of LQMFN further enhances the model's performance. Through amplitude encoding, LQMFN efficiently handles high-dimensional multi-modal data, and through quantum convolution and quantum pooling operations, it achieves deep fusion of complex features, such as text, images, and propagation network structure. Experimental results show that LQMF-RD not only achieves significant advantages in computational efficiency and detection performance, but also demonstrates strong robustness against quantum noise.

Although LQMF-RD demonstrates excellent performance in experiments, its practical application still faces two major challenges: insufficient fault tolerance of quantum hardware and limited generalization ability across different scenarios. Additionally, real-time monitoring of social media introduces further implementation difficulties, such as the need to ensure low-latency processing under massive high-speed streaming data to achieve timely rumor detection. The limitations of quantum hardware resources and noise issues also increase the complexity of system deployment. Future research will focus on addressing these practical challenges by developing more lightweight quantum circuit architectures suitable for noisy intermediate-scale quantum (NISQ) devices, exploring variational quantum error correction techniques, and optimizing the computational efficiency and resource consumption of algorithms. Meanwhile, building cross-lingual and multi-platform rumor detection benchmarks combined with federated learning frameworks will facilitate effective model transfer across different scenarios. By integrating edge computing and distributed system architectures, the model's usability and stability in large-scale real-time environments can be further enhanced.

CRediT authorship contribution statement

Keliang Jia: Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization; **Fanxu Meng:** Writing – original draft, Validation, Software, Formal analysis; **Ziwen Chen:** Software, Data curation; **Mengyao Du:** Resources, Investigation; **Jing Liang:** Writing – review & editing, Visualization.

Data availability

We will share the link to our data/code with the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix A: Symbol Table

Table A.11
Symbol table for LQMF-RD.

Symbol	Definition	Description
$\|0\rangle, \|1\rangle$	Computational basis states	Two standard basis states of a single qubit.
$\|i\rangle$	i-th computational basis state	Qubit state combination encoded in binary form.
$\|\psi\rangle$	Quantum state	Complete state vector of the system, containing all probability amplitudes.
n	Number of qubits	The number of qubits required for encoding, satisfying $2^n \geq$ feature dimension.
x_i	i-th component of input vector	The i-th element of the multi-modal feature vector.
C	Normalization constant	$C = \sqrt{\sum_{i=1}^{2^n} \|x_i\|^2}$, ensuring the quantum state is normalized.
$R_y(\theta)$	Y-axis rotation gate	Rotates a qubit by angle θ around the Y-axis on the Bloch sphere.
$R_z(\theta)$	Z-axis rotation gate	Rotates a qubit by angle θ around the Z-axis on the Bloch sphere.
$\text{CNOT}_{c\to t}$	Controlled-NOT gate	Flips the target qubit t when the control qubit c is in state $\|1\rangle$.
$U_{\text{conv}}^{\langle p,q\rangle}(\theta)$	Quantum convolution operator	Convolution operation on qubits p, q, composed of rotation gates and CNOTs.
$U_{\text{pool}}^{\langle p,q\rangle}(\theta)$	Quantum pooling operator	Dimensionality reduction operation on qubits p, q.
N	Dimension of compressed features	Dimensionality of multi-modal features fed into LQMFN.
M	Measurement operator	Operator used to measure a specific qubit.
$E_i(\theta)$	Expectation value	Expected measurement result used for model prediction.
θ_k	k-th trainable parameter	Tunable rotation angle of a quantum gate.
U_e	Encoding unitary operator	Transformation mapping classical features into a quantum state.
$U_a(\theta)$	LQMFN unitary operator	Overall unitary transformation of the quantum fusion network.
$\otimes$	Tensor product	Composition of multi-qubit states.
H.c.	Hermitian conjugate	Complex conjugate transpose, used in gradient calculation.

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